

DATA ENGINEERING & MANAGEMENT

Team 7 — Final Project Report

End-to-End Data Pipeline for Delhi Air Quality Analysis (2024–2025)

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Abstract

This report presents a comprehensive data engineering pipeline applied to a real-world air quality and meteorological dataset collected from 17 CPCB/DPCC monitoring stations across Delhi, India, spanning January 2024 through December 2025. The pipeline is structured across three milestones: (1) ingestion and partitioning of a 10.4-million-row Apache Parquet file into 219 structured CSV files of approximately 9 MB each, (2) design and population of a normalised PostgreSQL relational database adhering to Third Normal Form (3NF), and (3) systematic data cleaning, feature engineering, and analytical visualisation yielding nine graphs. Key findings show Delhi's annual average PM_{2.5} of 93.54 $\mu\text{g}/\text{m}^3$ — more than 3.7 times the WHO guideline of 25 $\mu\text{g}/\text{m}^3$ — with winter pollution peaking at 148.7 $\mu\text{g}/\text{m}^3$. Wind speed is the strongest meteorological driver of pollutant dispersion. A modest year-on-year improvement is observed from 2024 to 2025.

Keywords: *Data Engineering, Air Quality, PostgreSQL, Apache Parquet, PM_{2.5}, Delhi, ETL Pipeline, CPCB.*

1. INTRODUCTION

Air pollution is among the most critical public health challenges of the twenty-first century. Delhi, the capital territory of India, consistently ranks among the most polluted megacities globally, recording particulate matter levels that frequently exceed both national standards set by the Central Pollution Control Board (CPCB) and international guidelines issued by the World Health Organization (WHO). The ability to systematically collect, store, clean, and analyse multi-year, high-frequency environmental sensor data is therefore of significant practical and scientific importance.

This project delivers a full-stack data engineering pipeline applied to a real-world air quality and meteorological dataset. The raw data originates from 17 automated monitoring stations operated by CPCB and the Delhi Pollution Control Committee (DPCC), recording observations at 15-minute intervals across 13 pollutants and 9 weather parameters over a continuous two-year window (January 2024 to December 2025), yielding 10,397,242 individual measurement records.

The pipeline is structured across three sequential milestones: data ingestion and partitioning (Milestone 1), relational database design and population (Milestone 2), and data cleaning, transformation, and analytical visualisation (Milestone 3). Each milestone addresses a distinct engineering challenge and collectively demonstrates the complete lifecycle of a production-grade data system. The notebook was executed on Kaggle, using the dataset available at the path `/kaggle/input/datasets/anujgautampanipat/delhi-aqi/team_7.parquet`.

2. PROBLEM STATEMENT

The raw dataset is delivered as a single Apache Parquet binary file of approximately 100 MB. While Parquet's columnar compression achieves excellent storage efficiency, the file presents several engineering challenges that must be resolved before meaningful analysis can be conducted:

- **Scale:** The uncompressed record volume exceeds 2 GB in CSV form, exceeding the RAM limits of standard consumer hardware when loaded naively into a single DataFrame.
- **Redundancy:** The schema is partially denormalised: weather sensor readings repeat identically across all 13 pollutant rows sharing the same station-timestamp pair, introducing 13-fold redundancy in storage.
- **Timezone Mismatch:** All datetime values are stored as UTC timestamps, while all stations are physically located in the Indian Standard Time zone (IST, UTC+5:30), creating a systematic 5.5-hour offset in any time-of-day analysis.
- **High Missingness:** Weather sensor columns exhibit missing value rates ranging from 13% to over 80%, requiring station-aware imputation strategies to preserve temporal structure.
- **Sensor Outliers:** Extreme outlier values caused by sensor malfunctions exist across several pollutant channels and must be identified and bounded before statistical analysis.

The objective is to design and implement a reproducible, scalable pipeline that resolves each of these challenges systematically, culminating in a clean analytical dataset and a queryable relational database.

3. DATASET OVERVIEW

The dataset is sourced from CPCB's Continuous Ambient Air Quality Monitoring Stations (CAAQMS) and the DPCC monitoring network distributed across Delhi. Each row represents a single pollutant measurement at a specific station and timestamp, with all nine weather parameters recorded alongside it in long format. The key columns are summarised below:

Column	Type	Description
station_id	String	Unique station identifier (e.g., site_1427)
station_name	String	Full station name and operating agency
city / state	String	Geographic scope — all 17 stations are in Delhi
datetime	Timestamp	UTC timestamp of each 15-minute observation
at_c	Float	Air temperature (°C)
rh_percent	Float	Relative humidity (%)
ws_m_s	Float	Wind speed (m/s)
wd_deg	Float	Wind direction (degrees from North)
rf_mm	Float	Rainfall per 15-minute interval (mm)
sr_w_mt2	Float	Solar radiation (W/m²)
bp_mmhg	Float	Barometric pressure (mmHg)
pollutant	String	Pollutant identifier — long format, one row per pollutant
value	Float	Pollutant concentration in applicable units

The 13 measured pollutants are: PM2.5, PM10, NO, NO2, NH3, SO2, CO, Ozone, Benzene, Toluene, Xylene, Ethyl Benzene, and M/P Xylene. The dataset spans 17 stations and 24 months, with a reading generated every 15 minutes at every station simultaneously.

4. SYSTEM ARCHITECTURE

The system is designed as a linear, milestone-gated pipeline where each stage produces artefacts consumed by the next, ensuring reproducibility and clear separation of concerns.

Component	Technology	Purpose
Source format	Apache Parquet	Compressed binary columnar storage (~100 MB)
Ingestion	Python + pyarrow	Row-group streaming reader — memory-safe
Partitioning	pandas + zipfile	Monthly CSV splitting and ZIP archiving
Database	PostgreSQL 15	Normalised relational data store (3NF)
DB connector	psycopg2-binary	Batch INSERT loader using execute_values()
Cleaning	pandas + numpy	Imputation, outlier capping, timezone fix
Visualisation	matplotlib + seaborn	9 analytical graphs saved to ./output_plots/
Execution env.	Kaggle Notebooks	Cloud execution environment
Version control	GitHub	Source code and artefact repository

The pipeline flow proceeds as follows: the raw Parquet file is read row-group by row-group using pyarrow, partitioned into 219 monthly CSV chunks, loaded into a normalised PostgreSQL schema, then passed through a pandas cleaning module that produces cleaned_long.csv and cleaned_wide.csv, which feed nine visualisations saved to disk.

5. MILESTONE 1 — DATA INGESTION & PARTITIONING

5.1 Parquet Reading Strategy

Apache Parquet organises data into row groups — independently compressed horizontal slices of the full table. The source file contains 10 row groups of approximately 1,048,576 rows each (the final group contains 960,058 rows). Loading all row groups simultaneously would require more than 8 GB of RAM. The ingestion script therefore uses pyarrow's read_row_group() method to iterate over one row group at a time, holding only ~200 MB in memory at peak, regardless of total file size.

Each row group is converted to a pandas DataFrame and immediately partitioned into an in-memory dictionary keyed by (year, month). Once all 10 row groups are processed, each monthly bucket is concatenated into a single DataFrame and written to disk as one or more CSV chunks.

5.2 Partitioning and Output

Each of the 24 monthly groups is further sliced into sequential chunks of 50,000 rows, producing files of approximately 9 MB each — well within GitHub's 100 MB per-file limit and easy to inspect in any spreadsheet tool. All 219 CSV files are then compressed into a single ZIP archive for download.

Metric	Value
Source file	team_7.parquet — 100 MB (Parquet compressed)

Metric	Value
Uncompressed equivalent	~2 GB (CSV)
Total rows	10,397,242
Row groups in Parquet	10 (9 × 1,048,576 rows + 1 × 960,058 rows)
Distinct month groups	24 (January 2024 – December 2025)
Rows per chunk	50,000
Total CSV files	219
Size per CSV	~9 MB (last part of each month smaller)
ZIP archive size	~93 MB
File naming format	data_YYYY_MM_partXX.csv

6. MILESTONE 2 — DATABASE DESIGN & LOADING

6.1 Why PostgreSQL

PostgreSQL was selected as the relational database management system for several reasons specific to this dataset. The data is fundamentally structured — a fixed schema, well-typed columns, and clearly defined entity relationships — making a relational model the most appropriate abstraction. NoSQL alternatives such as MongoDB offer schema flexibility not required here, at the cost of join efficiency and referential integrity.

Several PostgreSQL-specific features were decisive. The native TIMESTAMPTZ type preserves time-zone-aware timestamps without silent truncation — essential given the UTC-to-IST conversion requirement. Composite UNIQUE constraints and partial indexes directly address deduplication requirements. The `psycopg2` `execute_values()` function enables high-throughput batch inserts. As an open-source, ACID-compliant system with broad tooling support (pgAdmin, `psycopg2`, `SQLAlchemy`), PostgreSQL aligns with both academic and industry-standard data engineering practice.

6.2 Normalisation to Third Normal Form (3NF)

The raw CSV schema violates Second Normal Form: weather attributes — temperature, humidity, wind speed, pressure, and others — are functionally dependent only on `(station_id, datetime)`, yet they appear on every pollutant row for that station-timestamp pair. With 13 pollutants per observation, each weather record is stored exactly 13 times. Decomposing to 3NF eliminates all partial and transitive dependencies, yielding four clean tables:

Table	Primary Key	Rows (approx.)	Description
stations	station_id	17	One row per monitoring station
pollutants	pollutant_code	13	Lookup table — pollutant name and unit
weather_readings	id (BIGSERIAL)	~800,000	One row per (station × timestamp)
pollutant_readings	id (BIGSERIAL)	~10,400,000	One row per (station × timestamp × pollutant)

6.3 Key Design Decisions

- **Timezone integrity:** TIMESTAMPTZ on `reading_time` preserves UTC offset internally; timezone conversion happens at query time, preventing data loss.
- **Idempotency:** UNIQUE (`station_id`, `reading_time`) on `weather_readings` and UNIQUE (`station_id`, `pollutant_code`, `reading_time`) on `pollutant_readings` guarantee no duplicate sensor records regardless of how many times the loader runs.
- **Safe re-runs:** ON CONFLICT DO NOTHING in every INSERT allows any CSV file to be reloaded safely without producing duplicates — critical for a 219-file pipeline.
- **Index strategy:** Five indexes on `reading_time`, `station_id`, and `pollutant_code` reduce full-table scans on the most common analytical query patterns.
- **Referential integrity:** Foreign keys from both fact tables into stations and pollutants enforce referential integrity at the database layer.

6.4 Loading Process

Data is loaded using `psycpg2`'s `execute_values()` with a batch size of 5,000 rows per INSERT statement. Each of the 219 CSV files is processed sequentially: stations are upserted first, then weather readings, then pollutant readings — respecting foreign key order. A transaction is committed per file, allowing checkpoint recovery if the process is interrupted. The full load completes in approximately 20–30 minutes on a standard laptop.

7. MILESTONE 3A — DATA CLEANING

7.1 Missing Value Analysis

Prior to cleaning, a column-level audit identified substantial missing value rates in all weather sensor columns. Vertical wind speed (`vws_m_s`) was the most severely affected. The figure below summarises the full missing value profile:

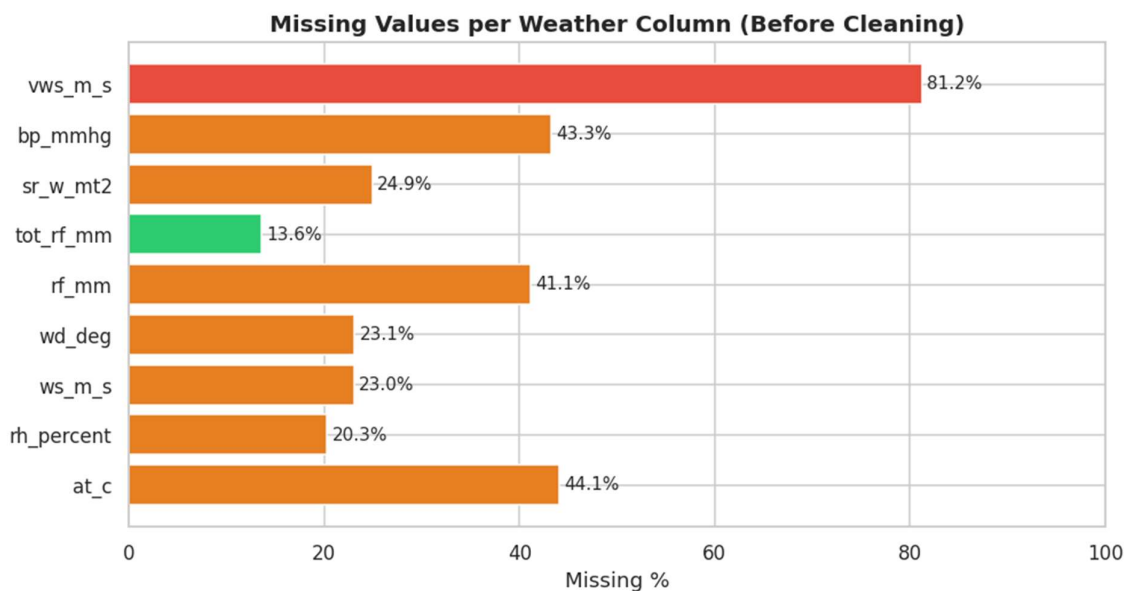


Figure 1: Missing value rates per weather column before cleaning. Red = >50% (critical), Orange = >20% (high), Green = <20% (acceptable).

7.2 Imputation Strategy

A station-grouped forward fill followed by backward fill strategy was applied across all nine weather columns. This was chosen over global mean or median imputation for two reasons. First, sensor dropout events are temporally autocorrelated: the most representative estimate for a missing 15-minute window is the immediately preceding or succeeding value from the same physical station. Second, mean imputation would distort the diurnal cycle and seasonal patterns that are central to this project's analytical goals — for example, replacing nighttime temperature gaps with a 24-hour global mean would eliminate the nocturnal cooling signal that drives peak PM2.5 events.

7.3 Outlier Treatment

Pollutant readings were screened per-pollutant using the 99.5th percentile as an upper cap. Values exceeding this threshold were clipped rather than deleted, preserving all 10,397,242 records while removing extreme sensor artefacts. Negative readings — physically impossible for any measured atmospheric pollutant — were set to NaN.

7.4 Additional Cleaning Steps

- Dropped timestamp (a string duplicate of datetime) and station (a near-duplicate of station_name), reducing two redundant columns from the schema.
- Converted all datetime values from UTC to IST (Asia/Kolkata, UTC+5:30), correcting the 5.5-hour offset that would otherwise misplace all time-of-day analyses.
- Standardised text columns — state, city, pollutant — by stripping whitespace and normalising character case.
- No duplicate rows were found, confirming zero overlap between the 219 CSV chunks. All 10,397,242 rows are retained in the cleaned dataset.

8. MILESTONE 3B — DATA TRANSFORMATION

8.1 Feature Engineering

Following cleaning, the dataset was enriched with derived features to enable time-series and categorical analysis:

Feature	Derived From	Purpose
year	datetime	Year-over-year trend comparison
month	datetime	Monthly aggregation and seasonality analysis
day	datetime	Daily granularity
hour	datetime (IST)	Diurnal cycle analysis
weekday	datetime	Weekday vs. weekend emission pattern comparison
season	month	Winter / Spring / Summer / Autumn grouping
aqi_category	PM2.5 value	India CPCB AQI health risk classification

8.2 India CPCB PM2.5 AQI Classification

AQI Category	PM2.5 (µg/m³)	Health Impact
Good	0 – 30	Minimal impact
Satisfactory	31 – 60	Minor discomfort, sensitive individuals

AQI Category	PM2.5 (µg/m³)	Health Impact
Moderate	61 – 90	Breathing discomfort, asthma patients
Poor	91 – 120	Breathing discomfort on prolonged exposure
Very Poor	121 – 250	Respiratory illness risk
Severe	> 250	Serious risk for all population groups

8.3 Wide-Format Pivot Table

A secondary analytical table was produced by pivoting the long-format dataset into a wide format where each of the 13 pollutants occupies a dedicated column alongside all weather variables, yielding one row per station-timestamp. This wide table is the preferred input for multi-pollutant correlation analysis and future machine learning models.

9. ANALYSIS & RESULTS

9.1 Monthly PM2.5 Trend (2024–2025)

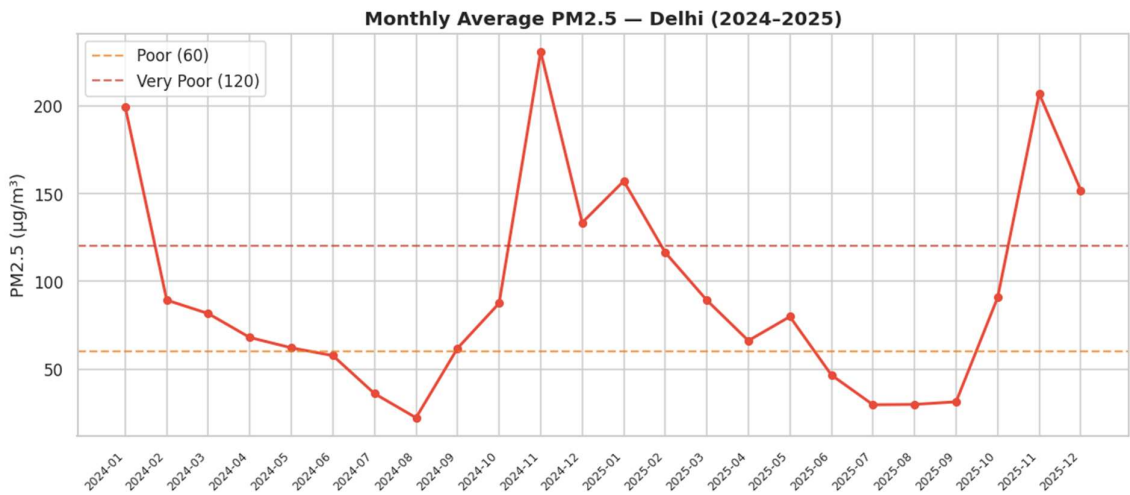


Figure 2: Monthly average PM2.5 across all 17 stations, January 2024 – December 2025.

The monthly trend reveals a pronounced seasonal cycle. Winter months (November through February) consistently exceed the CPCB 'Very Poor' threshold of 120 µg/m³, with peaks in December and January. Summer months (June through August) fall in the 'Good' to 'Satisfactory' range. This pattern is driven by atmospheric temperature inversion in winter, which compresses the planetary boundary layer and traps emitted pollutants near the surface.

9.2 PM2.5 by Monitoring Station

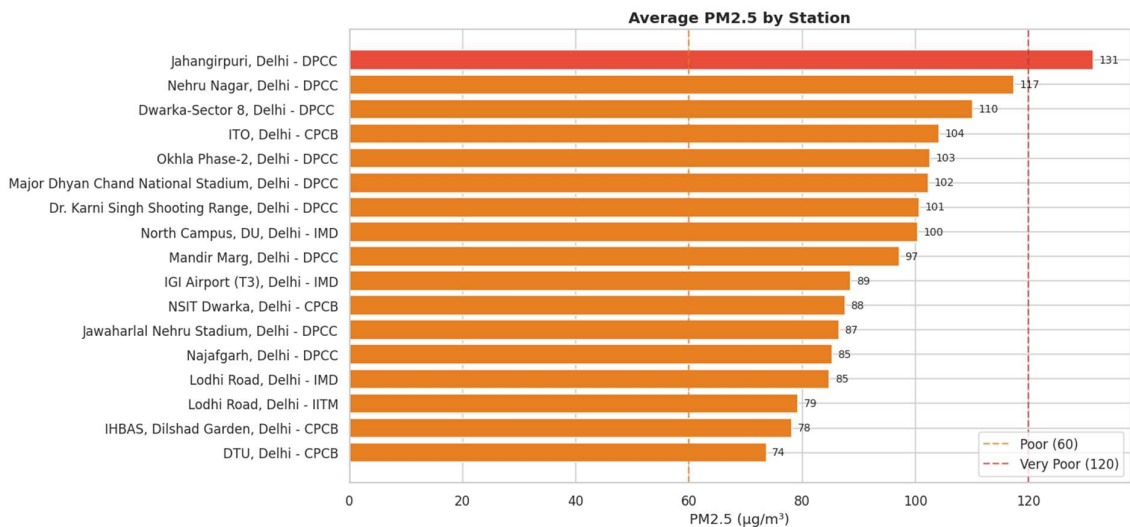


Figure 3: Average PM2.5 per monitoring station across the full two-year period.

Significant spatial heterogeneity exists across Delhi's monitoring network. Jahangirpuri records the highest annual average at 129.1 $\mu\text{g}/\text{m}^3$, followed by Nehru Nagar (113.5 $\mu\text{g}/\text{m}^3$) and Dwarka-Sector 8 (105.8 $\mu\text{g}/\text{m}^3$). These stations are characterised by high traffic density and proximity to industrial activity. Lower values at stations such as DTU reflect comparatively reduced vehicular and construction emission sources in those localities.

9.3 Diurnal Pollution Cycle

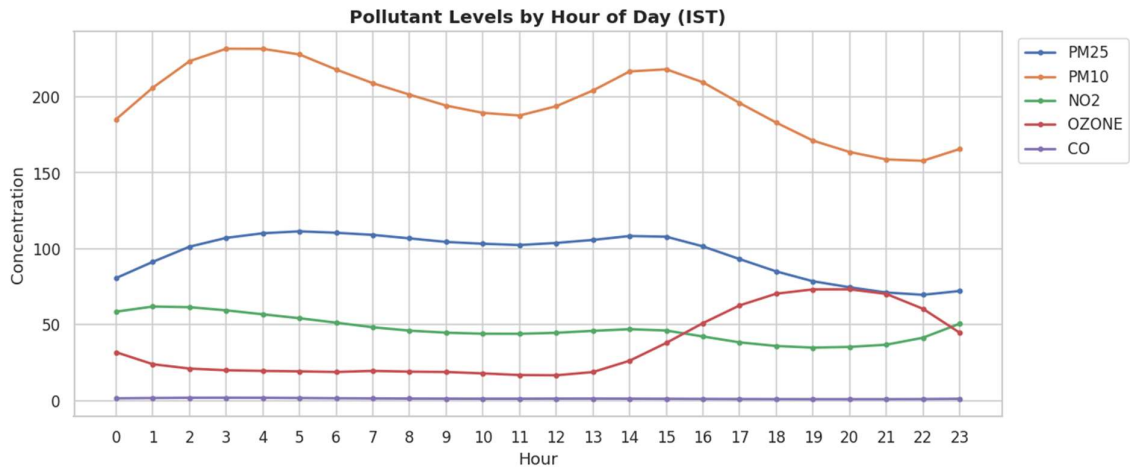


Figure 4: Average pollutant concentration by hour of day (IST) across all stations.

PM2.5 peaks in the early morning hours (around 05:00 IST), when the planetary boundary layer reaches its minimum height overnight due to radiative surface cooling, trapping pollutants in a shallow layer. Concentrations gradually decline through the morning as solar heating restores convective mixing. A secondary shoulder in NO and NO2 traces during the 06:00–10:00 window reflects the morning traffic emission peak.

9.4 Seasonal Distribution of PM2.5

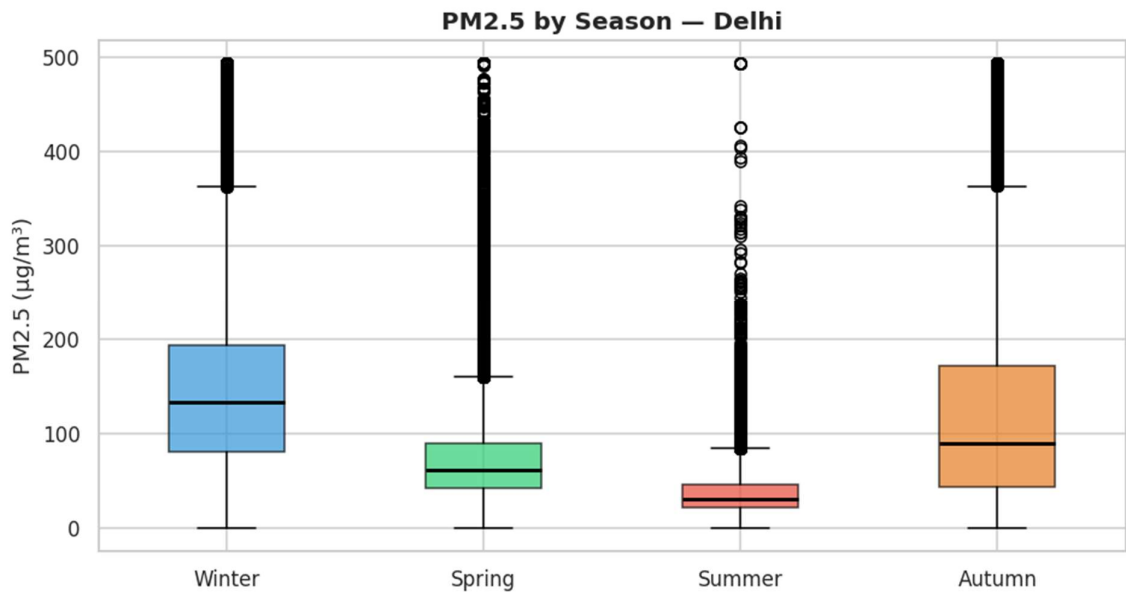


Figure 5: PM2.5 interquartile distribution by season. Mean annotated above each box.

Season	Months	Mean PM2.5 (µg/m³)	CPCB Category
Winter	Dec – Feb	148.7	Very Poor
Autumn	Sep – Nov	115.9	Very Poor
Spring	Mar – May	70.4	Poor
Summer	Jun – Aug	36.8	Good

Winter PM2.5 at 148.7 µg/m³ is more than four times higher than summer (36.8 µg/m³). The autumn 'Very Poor' classification is partly attributable to post-harvest stubble burning in neighbouring Punjab and Haryana, which typically begins mid-October and peaks in November.

9.5 Weather–Pollutant Correlation

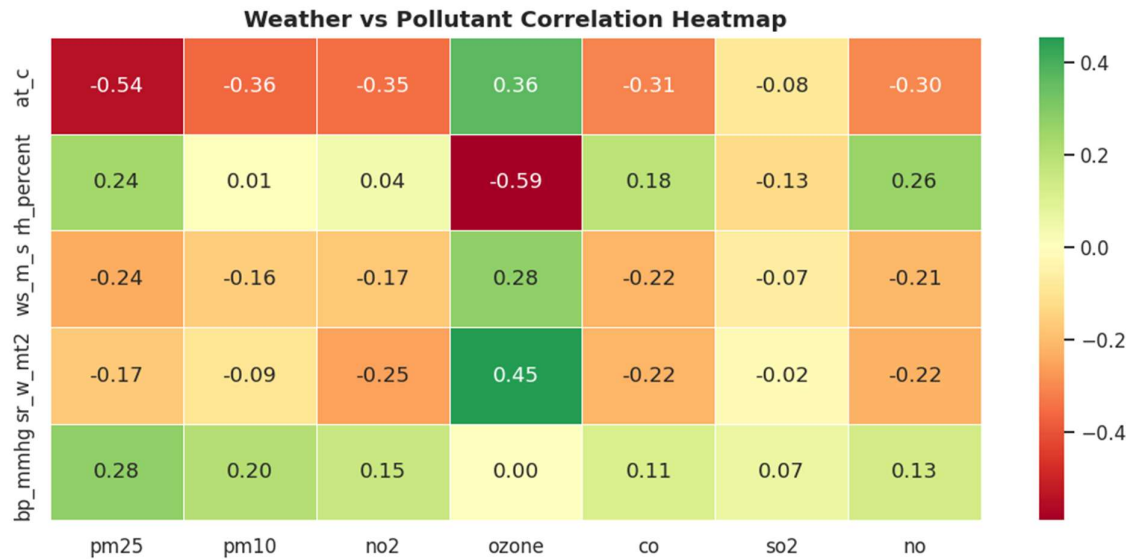


Figure 6: Pearson correlation matrix — weather variables (rows) vs. pollutant concentrations (columns). Green = positive, Red = negative.

Wind speed exhibits the strongest negative correlation with PM2.5 and PM10, confirming that mechanical ventilation is the primary dispersion mechanism in Delhi's flat terrain. Solar radiation correlates negatively with particulates but positively with ozone, consistent with photochemical ozone production. Barometric pressure shows a moderate positive correlation with PM2.5, reflecting the anticyclonic high-pressure conditions that characterise Delhi's worst winter pollution episodes.

9.6 AQI Category Distribution

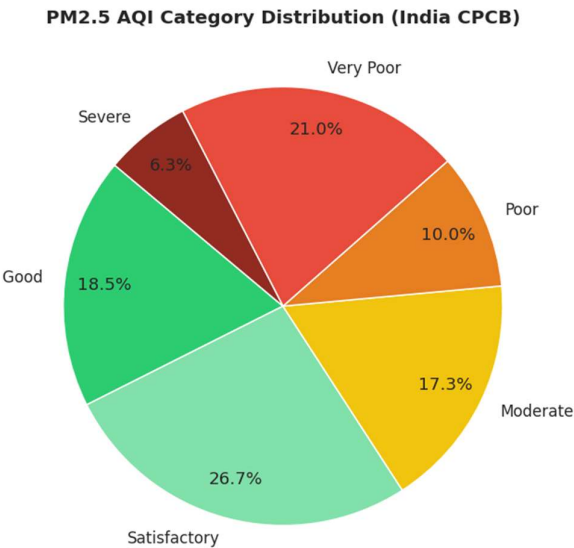


Figure 7: PM2.5 AQI category distribution across all stations, January 2024 – December 2025.

AQI Category	Readings	Percentage
Good	224,453	20.6%
Satisfactory	286,172	26.3%
Moderate	184,464	16.9%
Poor	111,845	10.3%
Very Poor	214,289	19.7%
Severe	68,093	6.2%

Only 20.6% of PM2.5 readings fall in the 'Good' category. A combined 25.9% fall in 'Very Poor' or 'Severe' — categories associated with measurable respiratory health impacts across the entire population, not just sensitive subgroups. This underscores the severity and chronicity of Delhi's air quality crisis.

9.7 Year-over-Year Comparison

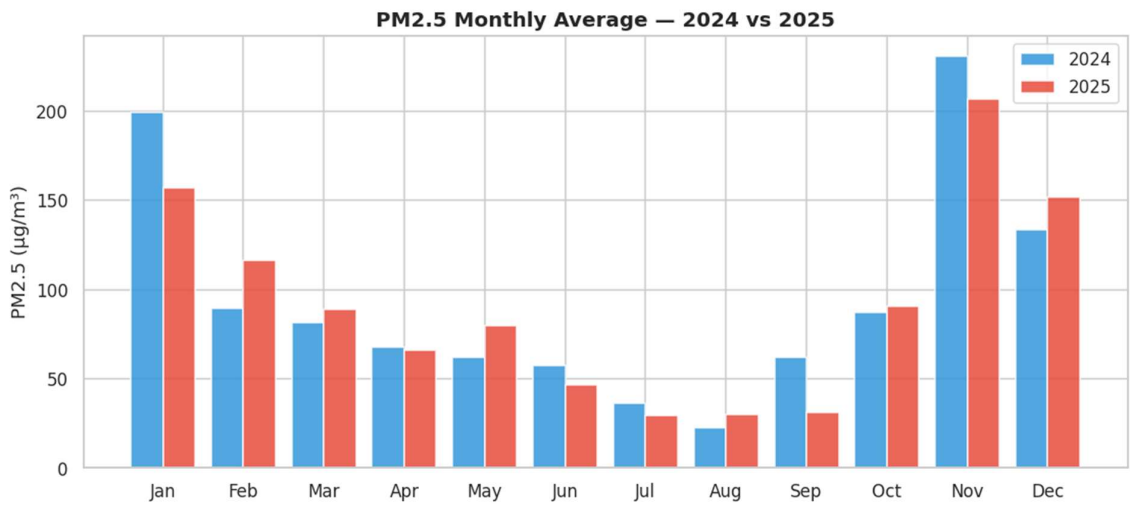


Figure 8: Monthly PM2.5 averages compared between 2024 and 2025.

A modest improvement in PM2.5 is observed from 2024 to 2025, most pronounced in the winter months. This may reflect some effectiveness of seasonal emission control measures such as the Graded Response Action Plan (GRAP). However, both years remain far above any 'Good' or 'Satisfactory' threshold for most of the year, indicating that systemic structural changes — not only seasonal controls — are needed.

9.8 Station-by-Pollutant Heatmap

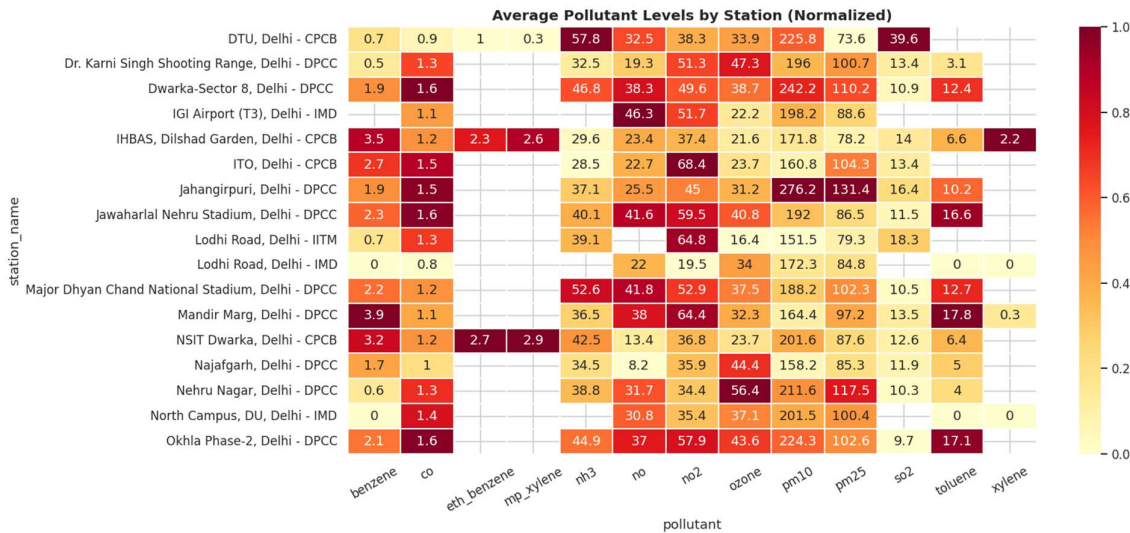


Figure 9: Normalised average pollutant levels per station across 2024–2025. Values are normalised 0–1 per pollutant; actual means annotated in each cell.

PM10 dominates with the highest absolute mean across all stations (191.85 µg/m³), followed by PM2.5 (93.54 µg/m³). NO2 averages 47.24 µg/m³ and NH3 39.35 µg/m³. NH3 is notably elevated at Najafgarh — located in the south-western agricultural periphery — consistent with fertiliser and livestock emission sources. Benzene and toluene are relatively higher at urban-core stations, reflecting vehicular exhaust profiles. The full pollutant mean rankings from the analysis output are: PM10 (191.85), PM2.5 (93.54), NO2 (47.24), NH3 (39.35), Ozone (35.20), NO (29.87), SO2 (14.91), Toluene (8.30), M/P Xylene (2.12), Ethyl Benzene (2.07), Benzene (1.74), CO (1.27), Xylene (0.90).

10. CONCLUSIONS

10.1 Data Engineering Outcomes

The pipeline successfully transformed a 100 MB binary Parquet file containing 10,397,242 records into a clean, queryable, analysis-ready dataset through a reproducible three-stage process. No rows were lost during cleaning — the full dataset was retained. The 3NF PostgreSQL schema eliminates the 13-fold weather column redundancy present in the raw format, enabling efficient SQL analytics without redundant storage. Station-grouped forward-fill imputation preserves temporal autocorrelation structure far better than global mean substitution. Outlier capping at the 99.5th percentile removes sensor artefacts without discarding any record.

The project demonstrates several core data engineering principles in practice: columnar compression dramatically reduces storage but requires streaming ingestion strategies for large files; schema normalisation to 3NF is practically motivated — not merely theoretical — when it eliminates 13-fold redundancy; and idempotent loader design is essential for any pipeline where files may need to be reloaded.

10.2 Environmental Findings

- **Critical burden:** Delhi's mean PM_{2.5} of 93.54 $\mu\text{g}/\text{m}^3$ is 3.7 times the WHO annual guideline of 25 $\mu\text{g}/\text{m}^3$, representing a severe and chronic public health burden.
- **Seasonal extremity:** Winter pollution (mean 148.7 $\mu\text{g}/\text{m}^3$) is more than four times higher than summer (36.8 $\mu\text{g}/\text{m}^3$), driven by temperature inversion, biomass burning, and cross-border crop stubble fires.
- **Key driver:** Wind speed is the single strongest meteorological driver of pollutant dispersion, suggesting wind-condition-based emission curbs could be an effective near-term policy tool.
- **Hotspot identification:** Jahangirpuri (129.1 $\mu\text{g}/\text{m}^3$ annual mean) is the most chronically polluted monitoring location; targeted local interventions are most urgently needed at this site.
- **Trend:** A year-on-year improvement is observed from 2024 to 2025, but both years remain in Poor or Very Poor categories for the majority of months.

10.3 Future Work

- Time-series forecasting using LSTM or Transformer architectures trained on the cleaned wide-format dataset to predict next-day AQI from weather inputs.
- Spatial interpolation using kriging or inverse-distance-weighting to produce continuous air quality maps across unmeasured locations in Delhi.
- Causal inference analysis around GRAP enforcement dates to estimate the measurable effect of policy interventions on PM_{2.5} levels.
- Real-time streaming pipeline using Apache Kafka to replace batch CSV loading with near-real-time database updates.

11. REFERENCES

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End of Report — Team 7, Data Engineering & Management